

Three Big Ideas Hiding Inside Days 13-15

ADVANCED OUTLOOK

Day 17 Enrichment · MathCraft · Grade 6 · Professor Xavier Honablué, M.Ed.

Advanced Outlook — not a required 6th-grade standard. Optional enrichment for curious students. Everything you do here is 6th-grade work: counting, area of triangles and trapezoids, and surface area (6.G.A.1, 6.G.A.3, 6.G.A.4). The ideas come from college-level geometry and topology. Nothing on this sheet will be on a test.

Big Idea 1 · Pick's Theorem: count dots, get area

On Day 13 and Day 15 you found areas of grid shapes by drawing a box and subtracting corners. In 1899 the Austrian mathematician Georg Pick found a shortcut that works for **any** polygon whose corners land on grid dots — no formulas for triangles or trapezoids needed at all. Count two things:

- **i** = the number of grid dots inside the shape (blue).
- **b** = the number of grid dots on the edges, corners included (orange).

Then the area is $A = i + b/2 - 1$. That is it — one formula for every straight-sided shape on a grid.

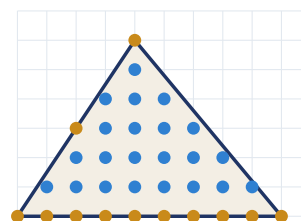
Test 1 — the Day 16 triangle, (0, 0), (9, 0), (4, 6).

Interior dots $i = 22$, boundary dots $b = 12$.

Pick: $A = 22 + 12 \div 2 - 1 = 22 + 6 - 1 = 27$.

Day 13 formula: $\frac{1}{2} \times 9 \times 6 = 27$. Same answer. ✓

● boundary (b) = 12 ● interior (i) = 22



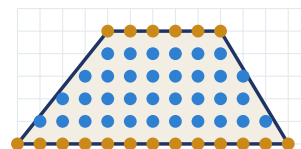
Test 2 — the Day 16 trapezoid, (0, 0), (12, 0), (9, 5), (4, 5).

Interior dots $i = 34$, boundary dots $b = 19$.

Pick: $A = 34 + 19 \div 2 - 1 = 34 + 9 - 1 = 42.5$.

Day 15 formula: $\frac{1}{2} \times (12 + 5) \times 5 = 42.5$. Same answer. ✓

● boundary (b) = 19 ● interior (i) = 34

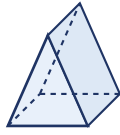
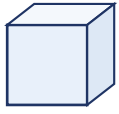


Your turn. On graph paper, draw any 5-sided shape with all corners on grid dots. Count i and b , use Pick's Theorem, then check by decomposing the shape into triangles and rectangles the Day 13 and Day 15 way. Try a shape with a dent in it (a concave one) — Pick's Theorem still works.

Why does the -1 belong there? Think of each boundary dot as "half owned" by the shape and half by the outside, which is where $b/2$ comes from. The extra -1 pays for the fact that going all the way around a closed shape turns a full 360° . A full proof is a favorite first project in college geometry courses.

Big Idea 2 · Euler's Formula: every solid keeps the same score

On Day 14 you unfolded prisms into nets. Count three things about any solid built from flat faces (a **polyhedron**): **V** = vertices (corners), **E** = edges, **F** = faces. Leonhard Euler discovered in 1750 that no matter which solid you pick, **$V - E + F = 2$** . Always.



cube: $8 - 12 + 6 = 2$ **triangular prism:** $6 - 9 + 5 = 2$

Solid	V	E	F	$V - E + F$
Cube	8	12	6	$8 - 12 + 6 = 2$
Rectangular prism (Day 14)	8	12	6	2
Triangular prism (Day 14)	6	9	5	$6 - 9 + 5 = 2$
Square pyramid	5	8	5	$5 - 8 + 5 = 2$
Tetrahedron (triangle-based pyramid)	4	6	4	$4 - 6 + 4 = 2$

Your turn. Build any prism or pyramid from a net (Day 14 skills), then count V, E, and F on your model and check Euler's score. Then try to break it: can you invent a solid where $V - E + F \neq 2$? (A solid with a hole through it, like a picture frame, gives 0 instead of 2 — that discovery started the branch of math called **topology**.)

Big Idea 3 · Surface area grows slower than volume

Day 14 measured the outside of a box. Here is what happens when the box gets bigger. For a cube with edge s : surface area is $6s^2$, volume is s^3 , and the ratio $SA : V$ is $6 \div s$ — it keeps shrinking as the cube grows.

Cube edge s	Surface area $6s^2$	Volume s^3	$SA \div V$
1 cm	6 sq cm	1 cu cm	6.0
2 cm	24 sq cm	8 cu cm	3.0
3 cm	54 sq cm	27 cu cm	2.0
6 cm	216 sq cm	216 cu cm	1.0
10 cm	600 sq cm	1,000 cu cm	0.6

Notice the 6 cm cube: surface area and volume are both 216, but they are not the same thing — one is square units and one is cubic units. That coincidence is a units trap worth arguing about.

Why it matters: crushed ice melts faster than one big block (more surface for the same amount of water). Small animals lose body heat faster than large ones. Engineers add fins to radiators and computer chips to buy extra surface area without extra volume. Architects of large buildings fight the opposite problem — a big building has relatively little skin to let light in.

Your turn. Compute $SA \div V$ for a 4 cm cube and a 5 cm cube. Then predict, without computing, whether a 20 cm cube's ratio is above or below 0.6 — and explain how you knew.

Answer Key

Big Idea 1: triangle — $i = 22$, $b = 12$, $A = 27$ square units, matching $\frac{1}{2} \times 9 \times 6 = 27$. Trapezoid — $i = 34$, $b = 19$, $A = 42.5$ square units, matching $\frac{1}{2} \times (12 + 5) \times 5 = 42.5$. Any student-drawn lattice polygon should match its decomposition answer exactly.

Big Idea 2: every prism and pyramid gives 2. A doughnut-shaped solid (one hole) gives 0; each extra hole subtracts another 2 — that number is called the Euler characteristic.

Big Idea 3: 4 cm cube $\rightarrow 96 \div 64 = 1.5$; 5 cm cube $\rightarrow 150 \div 125 = 1.2$; a 20 cm cube gives $6 \div 20 = 0.3$, which is below 0.6, because the ratio is always $6 \div s$ and s got bigger.

Explore More

MIT, Stanford, and Carnegie Mellon sources first. Everything here is optional.

MIT	MIT PRIMES Circle — free math enrichment reading program Advanced Outlook · Where ideas like Pick's Theorem get read and written up by students; a picture of where this path leads.
MIT	MIT BLOSSOMS — free video lessons Advanced Outlook · Browse the math lessons for geometry and measurement topics.
STANFORD	YouCubed at Stanford — open tasks 6th-grade friendly · Visual, low-floor/high-ceiling tasks; good partners for the dot-counting work.
CARNEGIE MELLON	CMU CS Academy — free Python graphics course Advanced Outlook · Code a lattice polygon and have the computer count the dots for you.
NRICH · CAMBRIDGE	NRICH (University of Cambridge) — Euler's Formula Advanced Outlook · A clear write-up of $V - E + F = 2$ and what happens to solids with holes.
MATHWORLD	Wolfram MathWorld — Pick's Theorem Advanced Outlook · The formal statement of $A = i + b/2 - 1$ and where it generalizes.
GEOGEBRA	GeoGebra — Net for a Rectangular Prism 6th-grade friendly · Fold and unfold a prism while counting faces, edges, and vertices.